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Iron and Manganese toxicity

Iron and Manganese exhibit competitive behaviour. Deficiency of Fe and Mn shows similar symptoms. Iron toxicity will affect absorption of manganese. The possible reason for iron toxicity is excess usage of chelated iron in addition with increased acidity of soil (PH less than 5.8) Iron and manganese toxicity will be solved by using fertilizer with balanced ratio of Fe and Mn.

inhibition of cell division and binding of plasma membrane with Calmodulin.

For theories regarding, translocation of minerals please refer Chapter- 11.

12.6 Hydroponics and Aeroponics

1. **Hydroponics or Soilless culture:** **Von Sachs** developed a method of growing plants in nutrient solution. The commonly used nutrient solutions are **Knop solution** (1865) and **Arnon** and **Hoagland Solution** (1940). Later the term Hydroponics was coined by **Goerick** (1940) and he also introduced commercial techniques for hydroponics. In hydroponics roots are immersed in the solution containing nutrients and air is supplied with help of tube (Figure 12.3).

Aeroponics: This technique was developed by **Soifer Hillel** and **David Durger**. It is a system where roots are suspended in air and nutrients are sprayed over the roots by a motor driven rotor (Figure 12.4).

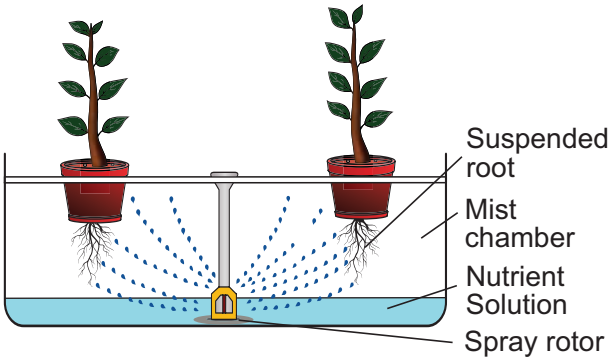


Figure 12.4: Aeroponics

12.7 Nitrogen Fixation

Inspiring act of nature is self-regulation. As all living organisms act as tools for bio geo chemical cycles, nitrogen cycle is highly regulated. Life on earth depends on nitrogen cycle. Nitrogen occurs in atmosphere in the form of N_2 (N≡N), two nitrogen atoms joined together by strong

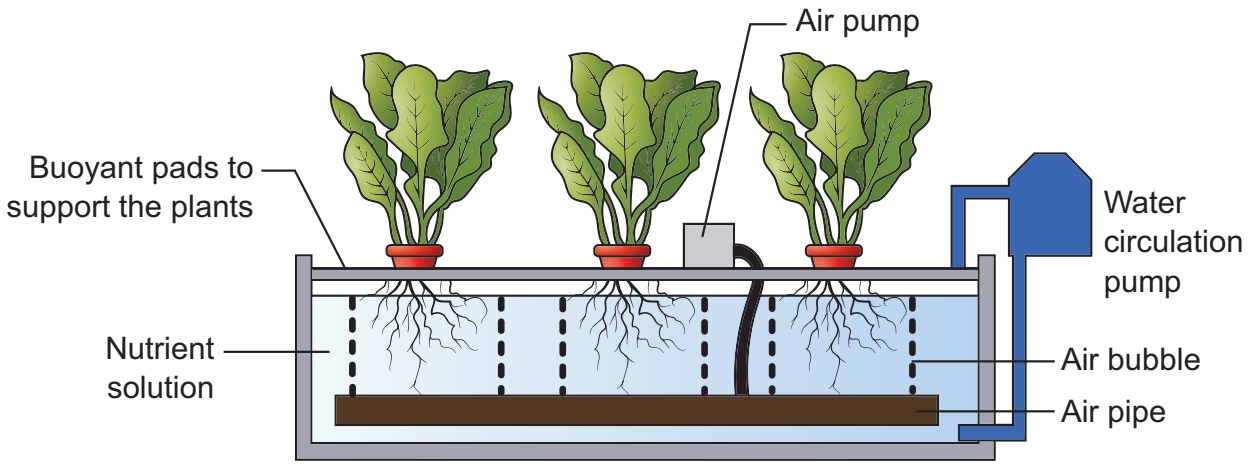


Figure 12.3: Hydroponics